

Ultra-Mek, Inc. v. Jiangsu Carya Smart Home Hardware Co., Ltd.

A public complaint analysis reports allegations concerning powered rocker-recliner and glider-recliner seating units and mechanisms. The issues described include mechanical linkage arrangements for reclining and reciprocating motion, power actuation, and product configuration. Separately, the Federal Register confirms that the same complainants filed an August 24, 2026 Section 337 complaint concerning certain powered rocker-recliner and glider-recliner mechanisms and seating units, requesting exclusionary and cease-and-desist relief against identified respondents.

Independent preliminary research based on public information.

FILED	FORUM	DOCKET	MATTER
2026-08-24	U.S. District Court for the Central District of California	2:26-cv-09471	Patent Infringement

Prepared 2026-08-27. No attorney or expert contacted. Availability, interest, compensation, and conflicts not checked.

Public filing, likely recipient, and technical frame

CAPTION	Ultra-Mek, Inc. et al. v. Jiangsu Carya Smart Home Hardware Co., Ltd. et al.
FORUM / DOCKET	U.S. District Court for the Central District of California / 2:26-cv-09471
FILED / TYPE	2026-08-24 / patent infringement
PUBLIC DOCKET	Open docket

VERIFIED PUBLIC RECORD

The public docket identifies the three plaintiffs, the named defendants, the August 24, 2026 filing date, C.D. Cal. docket number 2:26-cv-09471, a patent complaint under 35 U.S.C. § 271, and Alton Luther Absher III as counsel added for the plaintiffs. Its complaint entry identifies exhibits for U.S. Patent Nos. 8,016,348, 8,297,693, 8,398,165, and 8,398,168.

LIKELY RESEARCH PURCHASER

Alton L. Absher III - Outside counsel, Kilpatrick Townsend & Stockton LLP. The current public C.D. Cal. docket identifies Absher as the attorney added for the plaintiffs on the complaint. Treating the complaint filer as a potential purchaser of expert research is an analytical inference, not a statement of purchasing authority.

aabsher@ktslaw.com - publicly printed in the [linked professional source](#); not inferred.

TECHNICAL FRAME

A public complaint analysis reports allegations concerning powered rocker-recliner and glider-recliner seating units and mechanisms. The issues described include mechanical linkage arrangements for reclining and reciprocating motion, power actuation, and product configuration. Separately, the Federal Register confirms that the same complainants filed an August 24, 2026 Section 337 complaint concerning certain powered rocker-recliner and glider-recliner mechanisms and seating units, requesting exclusionary and cease-and-desist relief against identified respondents.

Secondary-source limitation: The complaint, asserted claims, accused-product identification, exhibits, inspection, CAD, design records, testing, and discovery must control. This preliminary issue map does not make an infringement, validity, damages, domestic-industry, or public-interest conclusion. [Open public synopsis](#)

REPORTED ASSERTED PATENTS

[U.S. Patent No. 8,016,348](#) | [U.S. Patent No. 8,297,693](#) | [U.S. Patent No. 8,398,165](#) | [U.S. Patent No. 8,398,168](#)

Questions likely to require expert input

This issue map is a research plan, not a legal conclusion. The filed complaint, patent exhibits, intrinsic record, and discovery must control.

01 Mechanism and linkage architecture

What pivots, links, brackets, frames, and mounting relationships are present in each accused rocker-recliner or glider-recliner mechanism, and do they correspond to the asserted claim limitations?

Potential evidence: Claim charts, CAD, engineering drawings, product samples, teardown photographs, bills of materials, assembly instructions, and dimensional inspection.

02 Reciprocating, rocking, and gliding motion

How does the seating unit move relative to its base during recline, rocking, or gliding, and what kinematic constraints and ranges of motion are created by the relevant linkage geometry?

Potential evidence: Physical testing, high-speed video, motion capture, kinematic models, CAD motion studies, drawings, and product demonstration records.

03 Power actuation and force path

What linear actuators, motors, transmissions, switches, and control logic move the seat, backrest, and footrest, and how is force transmitted through the claimed mechanism?

Potential evidence: Actuator specifications, wiring diagrams, firmware or controller documentation, force-displacement testing, service manuals, and component supplier records.

04 Component attribution and product versions

Which defendant, supplier, mechanism revision, and seating-unit model is associated with each alleged feature, and do versions materially differ?

Potential evidence: Revision histories, purchase orders, supplier drawings, model lists, product manuals, import and sales records, and sample provenance.

Questions likely to require expert input

05 Structural performance, fatigue, and safety

Do the particular joints, fasteners, stamped or welded components, and load paths provide the claimed functional performance under representative seating loads and cycles?

Potential evidence: Material specifications, finite-element models, fatigue and cycle tests, load cases, inspection records, certification tests, and failure analysis.

06 Prior-art and ordinary-skill baseline

What was known about powered reclining, rocker-recliner, glider-recliner, and linkage designs at the relevant priority dates, and how would a person of ordinary skill have understood the asserted terminology?

Potential evidence: Contemporaneous patents, catalogs, standards, technical texts, product manuals, engineering publications, and supported POSITA analysis.

07 ITC domestic industry and public interest

For the related Section 337 complaint, what U.S. articles and technical activities practice the asserted patents, and what are the practical effects of potential exclusionary relief on replacement supply and consumers?

Potential evidence: Domestic product engineering records, manufacturing and assembly evidence, investment records, product specifications, capacity data, supply-chain documents, and market technical evidence.

Candidates 1-2 for further diligence

Ranked for preliminary research priority based on subject-matter fit, relevant public work, testimony, and visible diligence burden. This is not a retention recommendation.

01 J. Michael McCarthy, PhD

Distinguished Professor, Mechanical and Aerospace Engineering and Electrical Engineering and Computer Science, University of California, Irvine

TECHNICAL FIT

Direct fit for linkage design, kinematics, spatial motion, and CAD-based analysis of articulated seating mechanisms.

RELEVANT WORK

UCI identifies his research as design of mechanical systems, computer-aided design, and kinematic theory of spatial motion.

PUBLIC LITIGATION EXPERIENCE

No specific public testifying engagement is asserted from this bounded review.

POINTS FOR FURTHER DILIGENCE

Strongest on mechanism geometry and kinematics; a separate test or furniture-product specialist may be required for particular manufactured product versions.

[Source 1](#)

[Draft email](#) | jmmccart@uci.edu | [public email source](#)

02 Larry L. Howell, PhD

Professor of Mechanical Engineering, Brigham Young University

TECHNICAL FIT

Direct fit for mechanism design, joint behavior, flexible and compliant mechanisms, and prototype-to-product analysis.

RELEVANT WORK

BYU identifies Howell as leading Compliant Mechanisms Research Group work using mechanism-design principles for engineered systems; his peer-reviewed work includes revolute-joint and spring/bearing behavior.

PUBLIC LITIGATION EXPERIENCE

No specific public testifying engagement is asserted from this bounded review.

POINTS FOR FURTHER DILIGENCE

Compliant mechanisms are not necessarily the asserted furniture mechanisms; engagement should be limited to the linkage, joint, and motion questions actually presented.

[Source 1](#) | [Source 2](#)

[Draft email](#) | lhowell@byu.edu | [public email source](#)

Candidates 3-4 for further diligence

03 Spencer P. Magleby, PhD

Professor of Mechanical Engineering, Brigham Young University

TECHNICAL FIT

Fit for mechanism architecture, deployable and articulated structures, and engineering design validation.

RELEVANT WORK

BYU's Mechanical Engineering materials identify Magleby as a professor and co-leader with Howell of the Compliant Mechanisms Research Group; the current BYU magazine describes their engineering work on jointed deployable structures.

PUBLIC LITIGATION EXPERIENCE

No specific public testifying engagement is asserted from this bounded review.

POINTS FOR FURTHER DILIGENCE

The public work located is not specific to furniture mechanisms; product and industry relationships require diligence.

[Source 1](#) | [Source 2](#) | [Source 3](#)

[Draft email](#) | magleby@byu.edu | [public email source](#)

04 Shorya Awtar, ScD

Professor of Mechanical Engineering and Joseph E. Shigley Collegiate Professor of Engineering, University of Michigan

TECHNICAL FIT

Fit for machine and mechanism design, kinematics, flexible systems, actuation, and human-centric mechanical systems.

RELEVANT WORK

Michigan identifies Awtar's research in mechanical design, precision engineering, mechatronics, robotics, constraint-based design, parallel kinematics, flexure mechanisms, and electromagnetic actuators; his lab reports multiple patented or patent-pending inventions.

PUBLIC LITIGATION EXPERIENCE

No specific public testifying engagement is asserted from this bounded review.

POINTS FOR FURTHER DILIGENCE

His precision/mechatronics profile may be broader than motion-furniture manufacturing; case-specific product evidence must determine fit.

[Source 1](#) | [Source 2](#)

[Draft email](#) | awtar@umich.edu | [public email source](#)

Candidates 5-6 for further diligence

05 David L. Trumper, PhD

Professor of Mechanical Engineering, Massachusetts Institute of Technology

TECHNICAL FIT

Fit for electromechanical actuation, motion control, sensing, dynamics, and analysis of powered mechanism subsystems.

RELEVANT WORK

MIT identifies Trumper as a mechanical-engineering professor and his public lab materials describe work in precision motion control, electromechanics, magnetic bearings, and integrated sensor design.

PUBLIC LITIGATION EXPERIENCE

No specific public testifying engagement is asserted from this bounded review.

POINTS FOR FURTHER DILIGENCE

His fit is strongest where motor, actuator, controller, or force-path issues are material, not necessarily for conventional furniture-linkage construction alone.

[Source 1](#) | [Source 2](#)

[Draft email](#) | trumper@mit.edu | [public email source](#)

06 Chinedum E. Okwudire, PhD

Professor of Mechanical Engineering and Director, University of Michigan Advanced Manufacturing Institute

TECHNICAL FIT

Fit for dynamics, vibrations, motion-delivery systems, manufacturing automation, and explaining how engineering design choices affect manufactured mechanisms.

RELEVANT WORK

Michigan identifies Okwudire as Professor and UMAMI director, with research in manufacturing automation, precision motion systems, dynamics, controls, mechatronics, and smart manufacturing; it records ASME Fellow recognition.

PUBLIC LITIGATION EXPERIENCE

No specific public testifying engagement is asserted from this bounded review.

POINTS FOR FURTHER DILIGENCE

The manufacturing-automation profile is indirect for consumer seating; may be most useful for motion, production, and machine-dynamics issues.

[Source 1](#) | [Source 2](#)

[Draft email](#) | okwudire@umich.edu | [public email source](#)

Candidates 7-8 for further diligence

07 Jwo Pan, PhD

Professor Emeritus of Mechanical Engineering, University of Michigan

TECHNICAL FIT

Fit for fatigue, structural mechanics, fasteners and joints, finite-element analysis, and failure testing of linkage components.

RELEVANT WORK

Public University of Michigan material identifies Pan as a mechanical-engineering professor emeritus and research interests in fracture and failure, fatigue testing and modeling, mechanical joints, and deformation analysis; SAE material gives the same professional email.

PUBLIC LITIGATION EXPERIENCE

No specific public testifying engagement is asserted from this bounded review.

POINTS FOR FURTHER DILIGENCE

Emeritus status and an older contact publication require a current availability and affiliation check before retention.

[Source 1](#) | [Source 2](#)

[Draft email](#) | jwo@umich.edu | [public email source](#)

08 Christopher D. Rahn, PhD

J. Lee Everett Professor of Mechanical Engineering, Pennsylvania State University

TECHNICAL FIT

Fit for dynamics, controls, mechanisms, and powered mechanical-system response.

RELEVANT WORK

Penn State's current Mechanical Engineering directory lists Rahn as J. Lee Everett Professor of Mechanical Engineering. His inclusion in the Larson Transportation Institute directory provides a current institutional contact source relevant to dynamic engineered systems.

PUBLIC LITIGATION EXPERIENCE

No specific public testifying engagement is asserted from this bounded review.

POINTS FOR FURTHER DILIGENCE

The directory source establishes role and email but does not by itself establish furniture-specific expertise; a detailed CV and publications review is needed before retention.

[Source 1](#) | [Source 2](#)

[Draft email](#) | cdr10@psu.edu | [public email source](#)

Candidates 9-10 for further diligence

09 Matthew B. Parkinson, PhD

Director of The Learning Factory and Professor, Pennsylvania State University

TECHNICAL FIT

Fit for mechanical design, ergonomics, anthropometry, biomechanics, user-facing product design, and seat-design evaluation.

RELEVANT WORK

Penn State identifies Parkinson's interests as rigorous design methods for artifacts used by people, including anthropometry, biomechanics, ergonomics, motion modeling, optimization, and mechanical design. Its publication listing includes vehicle-seat design work and compliant-mechanism design work with Howell.

PUBLIC LITIGATION EXPERIENCE

No specific public testifying engagement is asserted from this bounded review.

POINTS FOR FURTHER DILIGENCE

More useful for ergonomic, function, and product-design questions than purely patent claim construction or detailed actuator reverse engineering.

[Source 1](#) | [Source 2](#)

[Draft email](#) | mbp11@psu.edu | [public email source](#)

10 Timothy W. Simpson, PhD

Professor Emeritus of Mechanical Engineering, Pennsylvania State University

TECHNICAL FIT

Fit for engineering design methods, product families, product platforms, manufacturing, and comparing functionally related product variants.

RELEVANT WORK

Penn State identifies Simpson as Professor Emeritus in Mechanical Engineering, with research areas in design and manufacturing and work on product-family and product-platform design.

PUBLIC LITIGATION EXPERIENCE

No specific public testifying engagement is asserted from this bounded review.

POINTS FOR FURTHER DILIGENCE

Emeritus status and product-platform focus make him a secondary candidate unless versioning, design alternatives, or product-family questions become central.

[Source 1](#) | [Source 2](#)

[Draft email](#) | tws8@psu.edu | [public email source](#)

Preliminary candidates; no conflicts or availability determination

- Run party, affiliate, counsel, inventor, funder, and technology conflicts.
- Confirm testimony history, exclusions, compensation, publications, patents, and deposition performance.
- Investigate licensing, grants, consulting, commercial interests, and prior technical positions.
- Confirm role fit, availability, and interest only after counsel authorizes outreach.

ONGOING MATTER-SPECIFIC RESEARCH

Flint Witness Research works in the post-filing window, turning pleadings and public technical records into issue maps, researched candidate slates, and diligence starting points as the case develops.

[Schedule a 15-minute conversation](#)

CORE SOURCES

Public docket: <https://dockets.justia.com/docket/california/cacdce/2%3A2026cv09471/1035490>

Public professional email source: <https://ai-lab.exparte.com/attorney/152165/alton-abscher-iii>

Public technical context source: <https://ai-lab.exparte.com/case/dct/cacd/2%3A26-cv-09471/doc/analysis/1>

LIMITATIONS

The complaint, asserted claims, accused-product identification, exhibits, inspection, CAD, design records, testing, and discovery must control. This preliminary issue map does not make an infringement, validity, damages, domestic-industry, or public-interest conclusion.

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